



Java OOPs interview - surrounding questions

Constructor:

1. How will you call parameter less constructor, write one eg
2. How will you call parameterized constructor, write one eg
3. Can we have private constructor
4. Can I override constructor
5. Can I overload constructor
6. Why we can't override constructor
7. How will you call parent class constructor, which doesn't have parameters
8. How will you call parent class constructor, which have parameters
9. How will you call same class constructor from other constructor
10. Will it give compile time error if we give return type to constructor
11. In the inheritance which class constructor will be executed first
12. Is the constructor inherited to the child class?
13. `public class A { private A() {} }` . what is the correct way to create object for this class
14. `public class A { public A(int i) {} }` . Does Compiler supply default constructor for this class?
15.

```
public class A
{
    public A(int i) {}
}
public class B extends A
{
    public B(int i) {}
}
```

Will the above code compile successfully?

Encapsulation:

1. Give me one scenario example where you have implemented encapsulation. Students must tell one project scenario to explain the encapsulation.
2. What is the use of encapsulation?
3. which oops feature do you use to achieve data security [or] data hiding
4. How will you hide data from the external world using encapsulation? Hint : Students must talk about bean class.
5. Which methods are used to achieve encapsulation. Hint : setter() getter()
6. What is bean class
7. Write setter and getter methods for variable `private int sno;`

Abstraction:

1. How will you achieve abstraction in java?
2. Give me one real time example (not java) of abstraction?
3. Can we achieve 100% abstraction using abstract class? If not, why?

Inheritance:

1. Can I inherit private variables of parent class in child class?
2. What is the use of inheritance?
3. Is the constructor inherited to child classes?
4. Are final methods inherited to child classes?
5. Can I call / access final methods from child class?
6. Can I override final methods in the child class?
7. How do you call a parent class constructor from a child class?
8. What is object class?
9. If i don't want to inherit my class, what should i do?
10. Can I inherit the final class?
11. Why java doesn't support multiple inheritance?
12. What is function ambiguity in java? Show one example
13. How will you achieve multiple inheritance in java? Show one eg?
14. How many types of inheritances are there in java?
15. Which type of inheritance is not supported in java?
16. What is the limitation or disadvantage of inheritance in java.
17. Trainer can ask any coding based question on inheritance find output

Polymorphism:

1. Give one example for static polymorphism
2. Give one example for dynamic polymorphism

Method overloading

1. If 2 methods have the same name, same parameters, but different return type, is it overloading or error?
2. Can i overload constructor
3. Can i overload static methods
4. Can i overload final methods
5. How will you call overloaded methods, eg.
6. Trainer can give any code and ask if it is overloading or not.

Method overriding:

1. Can i override private methods of parent class in child class
2. Can i override abstract methods in child class
3. Can i override parent class constructor in child class, why?

4. If I don't want to override a parent class method in child class, what should I do?
5. What is the difference between private method and final method (as we can't override both methods in child class)?
6. Can I override parent class static methods in child class?
7. How do you call parent class overridden method from child class method. eg.
8. Trainer can ask any overriding based code find output.

Static and non static:

1. How will you access static variables [or] static methods of a class? Show one eg
2. How will you access non static variables [or] non static methods of a class? Show one eg
3. Which is faster, static methods or instance methods. Why?
4. Why is the main() method static?
5. Can I access non-static variables inside a static method? how?
6. Trainer can ask any coding find output question

Abstract methods and abstract classes:

1. If I have 0 abstract methods in a class, can I make that class an abstract class?
2. Can I create an object for an abstract class?
3. How will you access methods (which have bodies) of an abstract class?
4. Can I have a constructor in abstract class?
5. When will you use an abstract class, give one scenario with code (same question can be asked as what is the use of abstract class)
6. When will you use an abstract method, give one scenario with code (same question can be asked as what is the use of abstract method)
7. Can I achieve complete or full abstraction with the help of abstract class? Why if not?
8. Can I make an abstract class as a final class?
9. Take a parent class with few abstract methods, ask student to inherit into a child class. Ask student to call the methods with object.
10. Trainer can ask any coding question on abstract method or class

Interface:

1. What is the use of interface (same question can be asked as when will you use interface) give one scenario example with code
2. Does interface support multiple inheritance, show code
3. What is the access specifier used for the members of interfaces?
4. Can we achieve complete or full abstraction with an interface? Why
5. Can I implement multiple interfaces in a class? eg
6. Give one real world example (not java) of interface?
7. Can one interface inherit other interface
8. How will you implement an interface in a class and how will you call methods.
9. Trainer can ask any coding question on interface

Access specifier:

1. Can I access protected variables in other packages?
2. Can I access private variables in other classes?
3. Which access specifier is used for data security?
4. Difference bw protected and default
5. Trainer can ask any coding question on access specifier

Exceptions:

1. What is checked exception, give one eg.
2. What is unchecked exception, give one eg.
3. If I don't want to handle an exception, what should I do. Show eg.
4. If I use throws, and if an exception occurs in the current method, then what will happen. Who is going to handle that exception?
5. What is the use of throw. eg
6. How will you create custom exception. eg.
7. What happens if an exception occurs in the finally block? Will the finally block be executed?
8. If we use system.exit(), then finally block will be executed or not?

Collections & generics:

1. What is the use of collections
2. difference between array and collections
3. In which version java collections are introduced
4. What is the difference between collections and generics
5. Why generics are safe
6. difference between linked list and array list which is best
7. In which case linkedlist is better
8. Dif bw vector and arraylist
9. Dif bw hashtable and map
10. dif bw list and set [or] arraylist vs set
11. dif bw set and map
12. dif bw array and list
13. Dif bw list and map
14. Trainer can ask any coding question

Memory, garbage collection, threads and miscellaneous:

1. How many parts are there in memory
2. What is stored in stack
3. What is stored in heap
4. What is garbage collector
5. What is the meaning of un referenced objects
6. What is hashcode

7. What is synchronization
8. What is thread priority
9. What is main thread
10. What is set priority
11. What is jre vs jdk
12. What is the use of stack heap
13. What is current version of java
14. What is clone()
15. what is serializable interface
16. == vs equals()